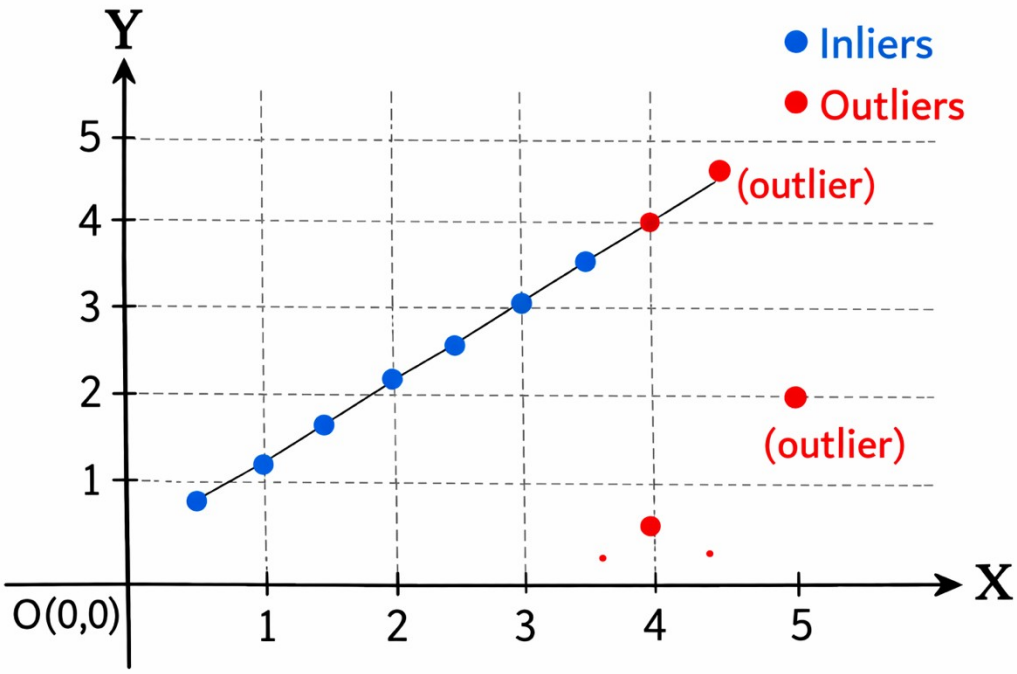
# Unit 3 - Session 3

## SLO 2: Implement RANSAC to handle outliers in line detection.

Duration: 10 Minutes Activity

**Question:**




No. The blue inlier points lie approximately on a straight line, while the red outliers do not.

**Q1.** From the given scatter of points, do all points appear to lie on a single straight line? Why or why not?

The outliers can significantly affect the fitted line, causing it to deviate from the main group of points.

**Q2.** If we apply **least-squares line fitting** to all the points shown, what problem is likely to occur?

A random sample is a randomly selected small set of data points used to estimate a model.

**Q3.** In the context of RANSAC, what is meant by a **random sample**?

2 points.

**Q4.** How many points are the **minimum required** to estimate a line model in RANSAC?

It measures the distance of each point from the selected line; points within the chosen tolerance are inliers.

**Q5.** After selecting a random line, how does RANSAC decide which points are **inliers**?

They are treated as outliers and are not used to support that model.

**Q6.** What happens to points that do not fit the selected line within the tolerance?

RANSAC repeatedly fits models using random samples and selects the model supported by the largest number of inliers, reducing the influence of outliers.

**Q7.** Why is RANSAC considered **robust to noise and outliers** compared to least-squares fitting?

The line model with the largest number of inliers (and the best fit among those) is chosen.

**Q8.** If RANSAC is run multiple times, which line model is finally chosen?